Odabrani koncepti relacionog modela podataka

Baze podataka 2 (2021/2022)
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Zatvarač skupa obeležja

- ZADATAK 1

  $U_1 = \{A, B, C, D, E\}$
  $F_1 = \{C \rightarrow D, BC \rightarrow A, B \rightarrow A, D \rightarrow E\}$

  $(AB)^+ = ?$

  $(CD)^+ = ?$

  $(ABC)^+ = ?$

  $(BC)^+ = ?$

Zatvarač skupa obeležja

- ZADATAK 1

  $U_1 = \{A, B, C, D, E\}$
  $F_1 = \{C \rightarrow D, BC \rightarrow A, B \rightarrow A, D \rightarrow E\}$

  $(AB)^+ = AB$

  $(CD)^+ = CDE$ [primena $D \rightarrow E$]

  $(ABC)^+ = ABCDE$ [primena $C \rightarrow D$, $D \rightarrow E$]

  $(BC)^+ = BCDAE$ [primena $C \rightarrow D$, $BC \rightarrow A$, $D \rightarrow E$]

Zatvarač skupa obeležja

- ZADATAK 2

  $U_2 = \{A, B, C, D, E, F, G, H\}$
  $F_2 = \{F \rightarrow C, DC \rightarrow E, H \rightarrow A, A \rightarrow H, BE \rightarrow C, E \rightarrow G\}$

  $(EF)^+ = ?$

  $(CD)^+ = ?$

  $(ABF)^+ = ?$

  $(BDF)^+ = ?$

Zatvarač skupa obeležja

- ZADATAK 2

  $U_2 = \{A, B, C, D, E, F, G, H\}$
  $F_2 = \{F \rightarrow C, DC \rightarrow E, H \rightarrow A, A \rightarrow H, BE \rightarrow C, E \rightarrow G\}$

  $(EF)^+ = EFGC$

  $(CD)^+ = CDEG$

  $(ABF)^+ = ABFCH$

  $(BDF)^+ = BDFCEG$

Zatvarač skupa obeležja

- ZADATAK 3

  $U_3 = \{A, B, C, D, E, F, G, H\}$
  $F_3 = \{B \rightarrow D, AC \rightarrow BD, E \rightarrow FC, FH \rightarrow A\}$

  $(EH)^+ = ?$

Zatvarač skupa obeležja

- ZADATAK 3

  $U_3 = \{A, B, C, D, E, F, G, H\}$
  $F_3 = \{B \rightarrow D, AC \rightarrow BD, E \rightarrow FC, FH \rightarrow A\}$

  $(EH)^+ = EHFCABD$

Ključevi šeme relacije

- ZADATAK 1

  $R_1 = \{A, B, C, D, E\}$
  $F_1 = \{D \rightarrow AC, C \rightarrow BE, A \rightarrow C\}$

  $K_1 = ?$

Ključevi šeme relacije

- ZADATAK 1

$R_1 = \{A, B, C, D, E\}$
$F_1 = \{D \rightarrow AC, C \rightarrow BE, A \rightarrow C\}$

$(ABCDE)^+ = ABCDE$	[uklanjanje A]
$(BCDE)^+ = BCDEA$	[uklanjanje B]
$(CDE)^+ = CDEAB$	[uklanjanje C]
$(DE)^+ = DEACB$	[uklanjanje D, E]
$E^+ = E$	!
$D^+ = DACBE$	D je ključ šeme relacije, nije moguća zamena

$K_1 = \{D\}$

Ključevi šeme relacije

- ZADATAK 2

  $R_2 = \{A, B, C, D, E, F\}$
  $F_2 = \{CE \rightarrow AB, F \rightarrow D, D \rightarrow F, B \rightarrow C, E \rightarrow C\}$

  $K_2 = ?$

Ključevi šeme relacije

- ZADATAK 2

$R_2 = \{A, B, C, D, E, F\}$
$F_2 = \{CE \rightarrow AB, F \rightarrow D, D \rightarrow F, B \rightarrow C, E \rightarrow C\}$

$(ABCDEF)^+ = ABCDEF$	[uklanjanje A]
$(BCDEF)^+ = BCDEFA$	[uklanjanje B]
$(CDEF)^+ = CDEFAB$	[uklanjanje C]
$(DEF)^+ = DEFCAB$	[uklanjanje D]
$(EF)^+ = EFDCAB$	[uklanjanje E, F] [EF je ključ]
$F^+ = FD$	!
$E^+ = ECAB$	!

Ključevi šeme relacije

- ZADATAK 2

$R_2 = \{A, B, C, D, E, F\}$
$F_2 = \{CE \rightarrow AB, F \rightarrow D, D \rightarrow F, B \rightarrow C, E \rightarrow C\}$

EF je ključ, moguća zamena nad EF

EF , $D \rightarrow F$ >> DE mogući ključ

DE [uklanjanje D, E] [DE je ključ]

$E^+ = ECAB$!

$D^+ = DF$!

Ključevi šeme relacije

- ZADATAK 2

  $R_2 = \{A, B, C, D, E, F\}$
  $F_2 = \{CE \rightarrow AB, F \rightarrow D, D \rightarrow F, B \rightarrow C, E \rightarrow C\}$

  dodatne zamene ne daju nove ključeve

  $DE , F \rightarrow D \;>>\; EF$

  $K_2 = \{EF, DE\}$

Ključevi šeme relacije

- ZADATAK 3

  $R_3 = \{A, B, C, D, E, F, G\}$
  $F_3 = \{F \rightarrow E, A \rightarrow B, C \rightarrow F, D \rightarrow A, E \rightarrow CFG\}$

  $K_3 = ?$

  B nije deo ključa
  B isključivo na desnim stranama FZ u F_3

  G nije deo ključa
  G isključivo na desnim stranama FZ u F_3

  D jeste deo ključa
  D isključivo na levim stranama FZ u F_3

Ključevi šeme relacije

- ZADATAK 3

  $R_3 = \{A, B, C, D, E, F, G\}$
  $F_3 = \{F \rightarrow E, A \rightarrow B, C \rightarrow F, D \rightarrow A, E \rightarrow CFG\}$

  $K_3 = ?$

  pokušaj sa D

  $D^+ = DAB$

  pokušaj sa dodavanjem F

  $(DF)^+ = DFEABCG$

  DF je ključ

Ključevi šeme relacije

- ZADATAK 3

  $R_3 = \{A, B, C, D, E, F, G\}$
  $F_3 = \{F \rightarrow E, A \rightarrow B, C \rightarrow F, D \rightarrow A, E \rightarrow CFG\}$

  $K_3 = ?$

  zamenom mogu biti dobijeni dodatni ključevi

  $DF , C \rightarrow F$ >> CD ...

  $DF , E \rightarrow F$ >> DE ...

  $K_3 = \{DF, CD, DE\}$

Važenje funkcionalne zavisnosti

- ZADATAK 1

  $F_1 = \{A \rightarrow B, B \rightarrow A, AB \rightarrow B, AB \rightarrow AB\}$

  $A \rightarrow B$?

  $B \rightarrow A$?

  $AB \rightarrow B$?

  $AB \rightarrow AB$?

relacija r_1

A	**B**
a1	b1
a2	b1
a1	b2
a3	b3

Važenje funkcionalne zavisnosti

- ZADATAK 1

  $F_1 = \{A \rightarrow B, B \rightarrow A, AB \rightarrow B, AB \rightarrow AB\}$

  $A \rightarrow B$ ne, uočiti (a1, b1) i (a1, b2)

  $B \rightarrow A$ ne, uočiti (a1, b1) i (a2, b1)

  $AB \rightarrow B$ da, trivijalna FZ

  $AB \rightarrow AB$ da, trivijalna FZ

relacija r_1

A	**B**
a1	b1
a2	b1
a1	b2
a3	b3

Važenje funkcionalne zavisnosti

- ZADATAK 2

$F_2 = \{C \rightarrow B, C \rightarrow D, AB \rightarrow C, AC \rightarrow B, D \rightarrow B, D \rightarrow C, D \rightarrow AC\}$

$C \rightarrow B$?

$C \rightarrow D$?

$AB \rightarrow C$?

$AC \rightarrow B$?

$D \rightarrow B$?

$D \rightarrow C$?

$D \rightarrow AC$?

relacija r_2

A	**B**	**C**	**D**
a1	b3	c1	d1
a2	b2	c1	d2
a3	b1	c2	d3
a3	b4	c2	d4

Važenje funkcionalne zavisnosti

- ZADATAK 2

$F_2 = \{C \rightarrow B, C \rightarrow D, AB \rightarrow C, AC \rightarrow B, D \rightarrow B, D \rightarrow C, D \rightarrow AC\}$

$C \rightarrow B$	ne
$C \rightarrow D$	ne
$AB \rightarrow C$	da
$AC \rightarrow B$	ne
$D \rightarrow B$	da
$D \rightarrow C$	da
$D \rightarrow AC$	da

relacija r_2

A	**B**	**C**	**D**
a1	b3	c1	d1
a2	b2	c1	d2
a3	b1	c2	d3
a3	b4	c2	d4

Implikacioni problem

- ZADATAK 1

  $U_1 = \{A, B, C, D\}$
  $F_1 = \{C \rightarrow AB, D \rightarrow B, D \rightarrow C\}$

  $F_1 \models D \rightarrow A$?

Implikacioni problem

- ZADATAK 1

  $U_1 = \{A, B, C, D\}$
  $F_1 = \{C \rightarrow AB, D \rightarrow B, D \rightarrow C\}$

  $F_1 \models D \rightarrow A$?

  1. način

  $D^+ = DBC\underline{A}$

  važi $F_1 \models D \rightarrow A$

Implikacioni problem

- ZADATAK 1

  $U_1 = \{A, B, C, D\}$
  $F_1 = \{C \to AB, D \to B, D \to C\}$

  $F_1 \models D \to A$?

  2. način

  $F_1 = \{C \to A, C \to B, D \to B, D \to C\}$

  $D \to C \mid$
  $C \to A \mid$ =(A3)=> $D \to A$

  važi $F_1 \models D \to A$

Implikacioni problem

- ZADATAK 2

  $U_2 = \{A, B, C, D, E\}$
  $F_2 = \{AC \rightarrow B, DB \rightarrow A, E \rightarrow A\}$

  $F_2 \models CDE \rightarrow A$?

Implikacioni problem

- ZADATAK 2

  $U_2 = \{A, B, C, D, E\}$
  $F_2 = \{AC \rightarrow B, DB \rightarrow A, E \rightarrow A\}$

  $F_2 \models CDE \rightarrow A$?

  1. način

  $(CDE)^+ = CDE\underline{A}B$

  važi $F_2 \models CDE \rightarrow A$

Implikacioni problem

- ZADATAK 2

  $U_2 = \{A, B, C, D, E\}$
  $F_2 = \{AC \rightarrow B, DB \rightarrow A, E \rightarrow A\}$

  $F_2 \models CDE \rightarrow A$?

  2. način

```
E  →  A   |
AC  →  B  |  =A3=>  CE  →  B  |
                    DB  →  A  |  =A3=>  CDE  →  A
```

  važi $F_2 \models CDE \rightarrow A$

Implikacioni problem

- ZADATAK 3

  $U_3 = \{A, B, C, D, E, F\}$
  $F_3 = \{CE \rightarrow D, F \rightarrow C, A \rightarrow F, AE \rightarrow C, B \rightarrow A\}$

  $F_3 \models BE \rightarrow D$?

Implikacioni problem

- ZADATAK 3

  $U_3 = \{A, B, C, D, E, F\}$
  $F_3 = \{CE \rightarrow D, F \rightarrow C, A \rightarrow F, AE \rightarrow C, B \rightarrow A\}$

  $F_3 \models BE \rightarrow D$?

  1. način

  $(BE)^+ = BEAFC\underline{D}$

  važi $F_3 \models BE \rightarrow D$

Implikacioni problem

- ZADATAK 3

  $U_3 = \{A, B, C, D, E, F\}$
  $F_3 = \{CE \rightarrow D, F \rightarrow C, A \rightarrow F, AE \rightarrow C, B \rightarrow A\}$

  $F_3 \models BE \rightarrow D$?

  2. način

  $B \rightarrow A$ |
  $E \in U_3$ | =A2=> $BE \rightarrow AE$ |
  $AE \rightarrow C$ | =A3=> $BE \rightarrow C$ |
  $CE \rightarrow D$ | =A3=> $BE \rightarrow D$

  važi $F_3 \models BE \rightarrow D$

Implikacioni problem

- ZADATAK 3

$U_3 = \{A, B, C, D, E, F\}$
$F_3 = \{CE \to D, F \to C, A \to F, AE \to C, B \to A\}$

$F_3 \models BE \to D$?

3. način

```
B → A |
A → F | =A3=> B → F |
               F → C | =A3=> B → C  |
                                E ∈ U3 | =A2=> BE → CE |
                                               CE → D  | =A3=> BE → D
```

važi $F_3 \models BE \to D$

Projekcija skupa FZ

- ZADATAK 1

  $U_1 = \{A, B, C, D\}$
  $F_1 = \{C \rightarrow A,\ B \rightarrow DC\}$

  $\pi_{|AC}(F_1) = ?$

Projekcija skupa FZ

- ZADATAK 1

$U_1 = \{A, B, C, D\}$
$F_1 = \{C \rightarrow A,\ B \rightarrow DC\}$

$\pi_{|AC}(F_1) = ?$

$A^+ = A$
$C^+ = CA$
$(AC)^+ = AC$

$$\begin{aligned}\pi_{|AC}(F_1) &= \{A \rightarrow A, A \rightarrow \emptyset, C \rightarrow AC, C \rightarrow \emptyset, AC \rightarrow AC,\ AC \rightarrow \emptyset, \emptyset \rightarrow \emptyset\} \\ &= \{A \rightarrow A, A \rightarrow \emptyset, C \rightarrow A, C \rightarrow C, C \rightarrow \emptyset, AC \rightarrow A, AC \rightarrow C, \\ &\quad AC \rightarrow \emptyset, \emptyset \rightarrow \emptyset\} \\ &= \{C \rightarrow A\} \cup \{\textit{Trivijalne FZ nad AC}\}\end{aligned}$$

Projekcija skupa FZ

- ZADATAK 1

  $U_1 = \{A, B, C, D\}$
  $F_1 = \{C \rightarrow A,\ B \rightarrow DC\}$

  $\pi_{|AB}(F_1) = ?$

Projekcija skupa FZ

- ZADATAK 1

$U_1 = \{A, B, C, D\}$
$F_1 = \{C \to A,\ B \to DC\}$

$\pi_{|AB}(F_1) = ?$

$A^+ = A$
$B^+ = B\cancel{DC}A$
$(AB)^+ = AB\cancel{DC}$

$\pi_{|AB}(F_1) = \{A \to A, A \to \emptyset, B \to AB, B \to \emptyset, AB \to AB,\ AB \to \emptyset, \emptyset \to \emptyset\}$
$= \{A \to A, A \to \emptyset, B \to A, B \to B, B \to \emptyset, AB \to A, AB \to B,$
$AB \to \emptyset, \emptyset \to \emptyset\}$
$= \{B \to A\} \cup \{\textit{Trivijalne FZ nad AB}\}$

Projekcija skupa FZ

- ZADATAK 2

  $U_2 = \{A, B, C, D, E, F\}$
  $F_2 = \{C \rightarrow D, B \rightarrow F, A \rightarrow EF, D \rightarrow C, FC \rightarrow E\}$

  $\pi_{|BC}(F_2) = ?$

Projekcija skupa FZ

- ZADATAK 2

$U_2 = \{A, B, C, D, E, F\}$
$F_2 = \{C \rightarrow D, B \rightarrow F, A \rightarrow EF, D \rightarrow C, FC \rightarrow E\}$

$\pi_{|BC}(F_2) = ?$

$B^+ = B\cancel{F}$
$C^+ = C\cancel{D}$
$(BC)^+ = BC\cancel{DFE}$

$\pi_{|BC}(F_2) = \{$*Trivijalne FZ nad BC*$\}$

Projekcija skupa FZ

- ZADATAK 2

  $U_2 = \{A, B, C, D, E, F\}$
  $F_2 = \{C \rightarrow D, B \rightarrow F, A \rightarrow EF, D \rightarrow C, FC \rightarrow E\}$

  $\pi_{|ADC}(F_2) = ?$

Projekcija skupa FZ

- ZADATAK 2

$U_2 = \{A, B, C, D, E, F\}$
$F_2 = \{C \rightarrow D, B \rightarrow F, A \rightarrow EF, D \rightarrow C, FC \rightarrow E\}$

$\pi_{|ADC}(F_2) = ?$

A+ = A~~EF~~
D+ = DC
C+ = CD
(AD)+ = AD~~EF~~C
(AC)+ = ACD~~EF~~
(DC)+ = DC
(ADC)+ = ADC~~EF~~

$\pi_{|ADC}(F_2) = \{D \rightarrow C, C \rightarrow D, AD \rightarrow C, AC \rightarrow D\} \cup \{$*Trivijalne FZ nad ADC*$\}$

Projekcija skupa FZ

- ZADATAK 2

  $U_2 = \{A, B, C, D, E, F\}$
  $F_2 = \{C \rightarrow D, B \rightarrow F, A \rightarrow EF, D \rightarrow C, FC \rightarrow E\}$

  $\pi_{|BDEF}(F_2) = ?$

Ekvivalencija dva skupa FZ

- ZADATAK 1

  $U = \{A, B, C, D\}$
  $F_1 = \{A \rightarrow B,\ B \rightarrow D, A \rightarrow C\}$
  $F_2 = \{A \rightarrow B, D \rightarrow A, A \rightarrow C\}$

  $F_1 \equiv F_2$?

Ekvivalencija dva skupa FZ

- ZADATAK 1

$U = \{A, B, C, D\}$
$F_1 = \{A \to B,\ B \to D, A \to C\}$
$F_2 = \{A \to B, D \to A, A \to C\}$

$F_1 \equiv F_2$?

$F_1 \models F_2$?

$F_1 \models A \to B$ važi

$A^+_{F_1} = A\underline{B}DC$

$F_1 \models D \to A$ ne važi

$D^+_{F_1} = D$

ne važi $F_1 \models F_2$

$\Rightarrow F_1 \not\equiv F_2$

Ekvivalencija dva skupa FZ

- ZADATAK 2

  $U = \{A, B, C, D, E\}$
  $F_1 = \{D \rightarrow C, E \rightarrow AC, C \rightarrow D, AB \rightarrow C\}$
  $F_2 = \{BE \rightarrow C, E \rightarrow D, E \rightarrow A, AB \rightarrow D\}$

  $F_1 \equiv F_2$?

Ekvivalencija dva skupa FZ

- ZADATAK 2

$U = \{A, B, C, D, E\}$

$F_1 = \{D \rightarrow C, E \rightarrow AC, C \rightarrow D, AB \rightarrow C\}$

$F_2 = \{BE \rightarrow C, E \rightarrow D, E \rightarrow A, AB \rightarrow D\}$

$F_1 \equiv F_2$?

$F_1 \models F_2$ važi

$F_1 \models BE \rightarrow C$ važi

$(BE)^+_{F_1} = BEA\underline{C}D$

$F_1 \models E \rightarrow D$ važi

$E^+_{F_1} = EAC\underline{D}$

$F_1 \models E \rightarrow A$ važi

$E^+_{F_1} = E\underline{A}CD$

$F_1 \models AB \rightarrow D$ važi

$(AB)^+_{F1} = ABC\underline{D}$

Ekvivalencija dva skupa FZ

- ZADATAK 2

$U = \{A, B, C, D, E\}$
$F_1 = \{D \rightarrow C, E \rightarrow AC, C \rightarrow D, AB \rightarrow C\}$
$F_2 = \{BE \rightarrow C, E \rightarrow D, E \rightarrow A, AB \rightarrow D\}$

$F_1 \equiv F_2$?

$F_2 \models F_1$?

$F_2 \models D \rightarrow C$ ne važi

$D^+_{F_2} = D$

ne važi $F_2 \models F_1$

$\Rightarrow F_1 \not\equiv F_2$

Literatura

- Pavle Mogin, Ivan Luković. Principi baza podataka. Fakultet tehničkih nauka i MP Stylos, Novi Sad. 1996.